1. More accurately, this appears to be a medieval conception with historical roots that go back to Aristotle.

2. As we will discuss later in the book, there are realms (such as the big bang and black holes) that still present many mysteries, at least in part owing to extremes of small size and huge densities that cause even Einstein's more refined theory to break down. So, the statement here applies to all but the extreme contexts in which the known laws themselves become suspect.

3. An early reader of this text, and one who, surprisingly, has a particular expertise in voodoo, has informed me that something *is* imagined to go from place to place to carry out the voodoo practitioner's intentions—namely, a spirit. So my example of a fanciful nonlocal process may, depending on your take on voodoo, be flawed. Nevertheless, the idea is clear.

4. To avoid any confusion, let me reemphasize at the outset that when I say, "The universe is not local," or "Something we do over here can be entwined with something over there," I am not referring to the ability to exert an instantaneous intentioned control over something distant. Instead, as will become clear, the effect I am referring to manifests itself as *correlations* between events taking place—usually, in the form of correlations between results of measurements—at distant locations (locations for which there would not be sufficient time for even light to travel from one to the other). Thus, I am referring to what physicists call *nonlocal correlations.* At first blush, such correlations may not strike you as particularly surprising. If someone sends you a box containing one member of a pair of gloves, and sends the other member of the pair to your friend thousands of miles away, there will be a correlation between the handedness of the glove each of you sees upon opening your respective box: if you see left, your friend will see right; if you see right, your friend will see left. And, clearly, nothing in these correlations is at all mysterious. But, as we will gradually describe, the correlations apparent in the quantum world seem to be of a very different character. It's as if you have a pair of "quantum gloves" in which each member can be either left-handed or right-handed, and commits to a definite handedness only when appropriately observed or interacted with. The weirdness arises because, although each glove seems to choose its